



# From Prototype to Product

The NPI journey — Rev A board to mass production

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NPI • DFM/DFA • Sourcing • Pilot • Ramp

Audience: Final-Year & Junior Engineers

Idea • Proto • Pilot • Production • Field



# Why the Gap is Wider Than You Think

A working prototype is ~10% of a shipped product

## Yield

Single proto works — but  
100 boards must work.

## Variation

Component spread, temperature,  
supply variation.

## Manufacturability

Hand-soldered prototype vs  
SMT line production.

## Quality

1 ppm defect rate target,  
incoming inspection.

## Compliance

EMC, safety, RoHS, BIS, FCC  
for every market.

## Field reliability

Failures appear at 1000-hour  
bathtub-curve edges.



# Design for Manufacturability & Assembly

How to design for the factory, not the bench

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- ▶ Single-sided assembly if possible (cheaper)
- ▶ Components only at recommended angles (0°/90°/180°/270°)
- ▶ Avoid components within 3 mm of board edge
- ▶ Same orientation for polarised parts
- ▶ Standard sizes (0603, SOIC, QFP) wherever possible
- ▶ Fiducials × 3 for panel + 3 per board
- ▶ Test points for ICT — every important net
- ▶ Programming header at edge for easy access
- ▶ Panelization with V-grooves or tabs
- ▶ BOM consolidation — fewer unique parts = cheaper
- ▶ Use parts on CM's preferred list
- ▶ Plan for assembly fixture (how it's positioned)

# Supply Chain & Sourcing

Critical for production, easy to get wrong

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- ▶ Avoid single-source parts — pick second-source alternates
- ▶ Authorized distributors only (DigiKey, Mouser, Arrow, Avnet) — counterfeit risk on grey market
- ▶ Check lifecycle status before designing in (Active vs NRND vs Obsolete)
- ▶ Lead time matters as much as cost — some parts have 52+ week lead
- ▶ Buy 10-20% extra inventory for losses & QA samples
- ▶ PCN (Product Change Notification) feeds — subscribe to vendor notifications
- ▶ Forecasting: share monthly volumes with vendor in advance for allocation
- ▶ Inventory management: cycle stock + safety stock + work-in-progress

# Production Documentation Package

What the CM needs to build your product

## Schematic PDF

Current rev with title block,  
signed by designer.

## Gerber + drill

RS-274X format,  
included ZIP file.

## BOM (.xlsx)

Ref-des, MPN, value,  
package, source, qty.

## Pick-and-Place

Centroid: X/Y/rotation/side  
for every SMD part.

## Assembly drawing

Layered view with notes,  
orientations highlighted.

## Functional spec

What does it do?  
How to test it works?

# Production Test Strategy

Catching defects before customers do

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- ▶ AOI (Automated Optical Inspection) — visual defects
- ▶ X-ray for BGAs — voids, shorts inside the package
- ▶ ICT (In-Circuit Test) — bed-of-nails fixture, every component
- ▶ FCT (Functional Test) — board powered up, firmware run, comms validated
- ▶ Boundary scan / JTAG — IC interconnect testing
- ▶ Burn-in — high-temp run for 24-48 hrs to catch infant mortality
- ▶ Final QA — calibration, identification, packaging check
- ▶ Statistical quality control: SPC charts, Cpk, defects-per-million

# Cost Engineering

Where the money goes

| Item                | Volume 1k   | Volume 100k | Notes                       |
|---------------------|-------------|-------------|-----------------------------|
| PCB                 | ₹100-300    | ₹30-80      | Layer count, finish         |
| BOM                 | ₹300-1500   | ₹150-700    | Negotiated at volume        |
| Assembly labor      | ₹100-300    | ₹20-50      | Per-board placement count   |
| Test                | ₹50-150     | ₹10-40      | ICT/FCT amortised           |
| Enclosure (plastic) | ₹100-500    | ₹30-150     | Injection mold tooling      |
| Packaging           | ₹20-100     | ₹10-50      | Cardboard + foam            |
| NRE (one-time)      | Stencil, FW | Same        | Mold tooling can be ₹5-50 L |

## Rule of thumb

Retail price  $\approx$  BOM cost  $\times$  3-5 (consumer) or  $\times$  8-10 (premium / B2B). Margin must cover R&D, marketing, support, returns.



# Pilot Run — The Critical Step

Catch problems before mass production

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- ▶ Build 100-500 units at the CM as a test run
- ▶ Same parts, same process as MP — just smaller batch
- ▶ Yield target: 95%+ first-pass yield (FPY)
- ▶ Failures get root-cause analysed — drives final design tweaks
- ▶ FAI (First Article Inspection) signed off by you AND the CM
- ▶ Validate test jigs, assembly steps, packaging
- ▶ Time pilot to allow at least one re-spin cycle if needed
- ▶ Often: visit the CM in person during pilot — relationships matter

# Common Pitfalls Going to Production

What I've seen kill products

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- ▶ Designing prototype on a part with 52-week lead time
- ▶ Hand-tuned firmware that breaks on production hardware variations
- ▶ Not specifying tolerances — '10  $\mu$ F' became 8  $\mu$ F X5R in production
- ▶ Skipping pilot — going straight to 10k unit MP — yield 60%
- ▶ No FCT firmware — can't tell good board from bad without bench debug
- ▶ Missing compliance testing — products held at customs
- ▶ Single source supplier goes EOL — no alternate
- ▶ Mechanical interference only discovered in plastic enclosure

# Quality Management Systems

Standards that buyers ask for

## ISO 9001

Generic quality management.  
Minimum for B2B.

## IATF 16949

Automotive — required  
by every car OEM.

## AS 9100

Aerospace — required  
for aero & defense.

## IPC-A-610

Solder-joint acceptability.  
Class 1/2/3.

## J-STD-001

Soldered electrical/electronic  
assemblies — workmanship.

## ESD S20.20

ESD control program —  
factory-wide standard.

# Production Ramp

From pilot to full volume

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- ▶ Week 0: pilot 500 units, debug, freeze design
- ▶ Week 4: ramp to 1k units / week, fix process issues
- ▶ Week 8: 5k units / week, optimise cycle time
- ▶ Week 12+: full volume — 10k+ / week
- ▶ Continuous improvement: SPC charts, RMA analysis, design refinements
- ▶ Plan for re-spins: 'Rev B' typically 6-12 months after launch
- ▶ ECN (Engineering Change Notice) process for every design change
- ▶ Maintain spare-parts inventory for 5-10 years after end-of-sale

# Key Takeaways

Key takeaways from this lecture

## NPI is a journey

Concept → EVT → DVT → PVT → MP.

## DFM mindset early

Design for the factory.

## Source carefully

Multi-source, check lifecycle.

## Test strategy = quality

AOI/ICT/FCT cover everything.

## Pilot before MP

100-500 units catch issues.

## Document obsessively

Schematic + BOM + .pos + drawing.

# Recommended Resources

Where to learn more

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- ▶ Andrew 'Bunnie' Huang — 'The Hardware Hacker' (lessons from manufacturing in Shenzhen)
- ▶ Tony Fadell — 'Build' (iPod, Nest founder on product development)
- ▶ Eric Reis — 'The Lean Startup' (MVP, build-measure-learn cycle)
- ▶ Karl Ulrich — 'Product Design and Development' (textbook)
- ▶ Hackaday — Bring a Hack series (real product war stories)
- ▶ PCBWay / JLCPCB blog posts on DFM
- ▶ EMSNow — EMS / contract manufacturing industry news
- ▶ Local contract manufacturer site visits — invaluable

# Pre-Production Review Checklist

Last sign-off before mass production

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- ▶ All compliance tests passed (EMC, safety, etc.)
- ▶ Pilot yield  $\geq 95\%$
- ▶ Test jig validated against known-good board
- ▶ BOM finalised + alternates approved
- ▶ Supply commitment for 6-12 months volume
- ▶ ECN process documented & approved
- ▶ Production firmware locked & version-controlled
- ▶ Packaging tested (drop, vibration, humidity)
- ▶ Labels and IDs printed per spec
- ▶ Compliance marks on board / enclosure
- ▶ Field-support plan documented
- ▶ Sales/marketing has spec sheets, photos, datasheet

# Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

## References & Further Learning

- ▶ Andrew Huang — The Hardware Hacker
- ▶ Tony Fadell — Build
- ▶ Eric Ries — The Lean Startup
- ▶ Karl Ulrich — Product Design and Development
- ▶ Hackaday — manufacturing & teardown blog
- ▶ EMSNow — contract manufacturing news
- ▶ JLCPCB / PCBWay DFM guides
- ▶ Local CM visits — Bengaluru / Pune / Chennai

Thank you • Build something this week.